

Date:04.05.2026

Department of Electronics and Communication Engineering

for Enhancing Student Employability for the Batch 2024-2028

I. Background and Rationale

Engineering education today extends beyond academic knowledge. Employers expect graduates to demonstrate analytical thinking, quantitative ability, problem-solving skills, programming competence where relevant, and effective communication. Most recruitment processes begin with screening tests assessing aptitude, logical reasoning, technical fundamentals, and coding skills before interviews.

While the curriculum provides strong disciplinary foundations, students benefit from early exposure to industry-style assessments. Periodic evaluation of employability competencies can significantly enhance placement readiness.

To support this, the Chennai Campus proposes CARE (Career Assessment for Readiness and Employability) Diagnostic Tests. Conducted at multiple stages of the undergraduate program, these assessments will help students gauge their preparedness, identify skill gaps, and progressively build the competencies required for successful placements.

II. Objectives of CARE Diagnostic Tests

The CARE Diagnostic Tests aim to achieve the following objectives:

Assess Career Preparedness Early: Evaluate students' readiness in areas commonly assessed during recruitment processes.

1. **Identify Skill Gaps:** Provide actionable insights into gaps in aptitude, programming, core subjects, or analytical thinking.
2. **Provide Structured Feedback to Students:** Help students understand where they currently stand and what improvements are required.
3. **Enable Targeted Training Interventions:** Use assessment insights to design training programs in aptitude, coding, technical subjects, communication, and problem-solving.
4. **Categorize Students Based on Preparedness Levels:** Group students into categories such as advanced learners, developing learners, and emerging learners for tailored training support.

Strengthen Placement Readiness: Ensure progressive development before students enter the placement cycle.

III. Structure and Timeline of CARE Diagnostic Tests

To monitor and enhance employability readiness, five CARE Diagnostic Tests will be conducted during the undergraduate program.

CARE Diagnostic Test	Semester	Timing
CARE Diagnostic Test – 1	End of Semester IV	Intermediate technical and analytical readiness evaluation
CARE Diagnostic Test - 2	End of Semester V	Pre-placement readiness assessment
CARE Diagnostic Test - 3	End of Semester VI	Final placement preparedness evaluation

IV. Materials Prepared for Placement Readiness:

Students are provided with carefully curated study resources, including a set of “Must-Do Coding Questions” covering key topics such as Arrays, Strings, Sorting, Searching, Dynamic Programming, Trees, Digit Manipulation, Graphs, Linked Lists, and Stacks & Queues. This collection consists of around 400 questions designed to strengthen problem-solving skills and coding proficiency.

V. Activity Sheets:

The Activity Sheets are designed to enhance students’ coding skills through structured practice. Each activity sheet consists of 20 coding questions, categorized into Easy, Medium, and Hard levels, to ensure progressive learning and skill development.

A window period of 4 days is provided for the completion of each activity sheet. The activity sheets are scheduled from May 04, 2026 to May 27, 2026, with a total of 6 activity sheets planned for the month.

Submission Guidelines

- All solutions must be compiled into a single PDF document
- The file name must be your Roll Number
- Programs included in the PDF should be clearly written and properly formatted
- Students may use any of the following programming languages: C, C++, Python, or Java

Any standard online coding platform can be used to write and test the programs

The sheets will be shared by the respective Class Advisors via a centralized drive link. Students are required to complete the assigned problems and submit their solutions through the submission link provided by the Class Advisor.

Activity Sheet No.	Topic	Deadline	Link
Activity Sheet 01	Arrays – Part 1	07.05.2026	
Activity Sheet 02	Arrays – Part 2	11.05.2026	Activity Sheet_02_Arrays_Part 02_Final Year.pdf
Activity Sheet 03	Strings	15.05.2026	Activity Sheet_03 - Strings_Final Year.pdf
Activity Sheet 04	Sorting & Searching	19.05.2026	Activity Sheet_04 - Sorting_Searching_FinalYear.pdf
Activity Sheet 05	Dynamic Programming	23.05.2026	Activity Sheet_05 - Dynamic Programing - Final Year.pdf
Activity Sheet 06	Digit Manipulation	27.05.2026	Activity Sheet_06_DigitManipulation_FinalYear.pdf

VI. Diagnostic Test Syllabus:

Paper II - Technical	
Algorithm Basics	Heuristic approach/ Brute Force technique –Optimistic Approaches - Greedy approach, Divide and Conquer approach, Dynamic Programming approach. Complexity Analysis: Time Complexity and Space Complexity. Problem Solving Approaches – Iterative Approach, Recursive Approach
Data Structure and Algorithms	
Basics- DS	Array – Linked List –Stack-Queue and its Applications
Sorting Techniques	Bubble sort, Insertion Sort-Selection Sort – Merge Sort- Quick Sort- Applications of Sorting and Searching Algorithms.
Searching Techniques	Linear and Binary Search and Applications of Searching Algorithms.
Trees	Binary Search Tree- Tree Traversals-AVL Trees and Applications of Trees.

Graphs	Graph Traversals – Breadth First Search (BFS), Depth First Search (DFS), Shortest Path Algorithms – Dijkstra's Algorithm– Minimum Spanning Tree - Prim's Algorithm– Kruskal's Algorithm and Applications of Graph Algorithms
Hashing	Hash Tables, Hash Functions, Collision Resolution Techniques- Open Addressing- Closed Addressing Techniques- and Application of Hashing.
OOPS	
OOPS Concepts	
Problem solving	
(Coding Practice)	Arrays – Strings- Digit Manipulation – Sorting Algorithms – Searching Algorithms – Dynamic Programming – Graph Traversal (BFS, DFS)

Evaluation Pattern

Evaluation Pattern							
Paper	Conducted By	Evaluation Component	Details	Date	Duration	Marks	Mode of Exam
Paper 2	Dept Faculty	Coding Rounds(2 Easy + 1 Medium)	Problem Solving using Programming skills (3 Questions)	18.06.2026	90 minutes	70	Online
		MCQ (Output Prediction + Debugging)	Technical Objective Questions(30 Questions)		30 minutes	30	
Total(VQL+ Coding + MCQ)						200	

CARE Team Members:

Sno	Name of the Member	Designation
1	Dr.Lakshmanan.S.A	Assistant Professor & Chairperson, Department of ECE
2	Dr.R.Annamalai	CARE Team Head/ Assistant professor(Selection Grade) & Program Chair - AIE AI&DS M.tech CSE
3	Dr.Amirthalakshmi.T.M	CARE Member /Assistant professor (Selection Grade)
4	Dr.Damodaran.V	CARE Member/ Assistant professor (Senior Grade)
5	Dr.Rahul.S.G	Placement Coordinator/ Assistant professor (Senior Grade)
6	Dr.Gayathri.M	Placement Coordinator / Assistant professor
7	Dr.Vaishale.R	Faculty Mentor
8	Dr. Thenmozhi	Faculty Mentor

Prepared By
CARE Team

Approved By
Principal